Corrections and Clarifications

FFGFT / T0-Time-Mass Duality

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Contents

.1 Register Table 3
.2 Note on R92 (self-correction) 9
.3 Currently Open Bridges 9

Preamble

This document is the ongoing correction register of the FFGFT corpus. It records *exclusively* two items per entry: **(1)** the correction or clarification in brief form and **(2)** which document already contains the implemented correction.

Elaborations, derivations, and status notes belong in the target documents or in the archive editions (Doc. 190-Archive-1: up to R85; Doc. 190-Archive-2: R86–R114).

Principle (append-only): Source documents are *not* revised. Every declared correction must be visible in an *active* document of the corpus — either in a new successor document containing the corrected version, or as an addendum at the end of the affected document. An entry that exists only in the archive but appears in no active document is not complete. The column "Implemented in" of the register table names the active document carrying the correction; it remains empty while that is still outstanding.

Entry types: K = Correction (content error), P = Precision (clarification, not an error), R = Register entry (revision of a previously booked statement).

.1 Register Table

Full elaborations: Doc. 190-Archive-1 (K1–K7, P1–P43, R1–R85), Doc. 190-Archive-2 (R86–R114, K-041).

No.	Concerns	Subject	Implemented in
K1	049 (HTML)	$\xi = \lambda_h^2 v^2/(16\pi^3 m_h^2)$; earlier version counted one 2-sphere too many	Doc. 310
K2	116 (Tab.)	$r_\tau = 25/9$	Docs. 016, 046, 116 (De+En)
K3	018 (Explorer)	$a_e = 2\pi^2/(f \cdot k)$	Doc. 018
K4	267, 268	Factor 3 = three observable spatial dimensions (k^2_{obs} from $n_{1,2,3}$, fourth direction = time unfolding)	Docs. 267, 268
K5	275 (script v3)	Three errors in v3; all fixed in v4 (11 June 2026)	Script v4
K6	004, 025, 030, 039, 041, 053, 061, 063, 068, 081	Redshift is achromatic: $1 + z = e^{\xi x}$ independent of λ	Doc. 280
K7	rsa/ (14 HTML pages)	Calculation path, labels and method text corrected; no warning boxes, pages describe themselves	rsa/ directly
P1	116, 158	Both correct; $0.001\,\%$ for external use	Docs. 116, 158 (addendum)
P2	several	$\hbar$ follows from ξ via $L_0 = \xi \cdot \ell_P$	corpus-wide
P3	173–176	Two parameters: ξ_0 (geom.) and ξ_{Higgs} (alg.)	Docs. 173–176
P4	180–182	L_0 = geometric fundamental length	Doc. 180

No.	Concerns	Subject	Implemented in
P5	several	Mass structure from ξ; $\sim 1\,\%$ agreement	corpus-wide
P6	116, 158	Koide only with numerical values (non-closure, Doc. 193)	Doc. 193
P7	005, 008, 019, 086, 189, 202	Scale structure from recursion $\mathscr{R}$	respective docs
P8	005, 008, 012, 014, 041, 060, 133, 141, 159, 181	Geometric scaling / fractal correction	respective docs
P9	005, 006, 042, 046	Columns "scaling class" + "SM correspondence"; quarks via $m = r\xi^p v$	respective docs
P10	009	$K = R_{pe}/(4/3)^7$: empirical anchor; factor 1.314 implicitly defined	Doc. 009
P11	025, 003, 133	$(1 - 275/4 \cdot \xi)$, $\Delta = 0.0002\,\%$; Docs. 003 and 133 use different models	respective docs
P12	022, 230	Two observables: cumulative (022) vs. elementary $\xi/(2\pi)$ (230)	Docs. 022, 230
P13	013, 025, 061	$\xi_{\text{FFGFT}}/\xi_{\text{UIFT}} \approx 414{,}684$; table in 013 reconstructed	Doc. 013
P14a	041, 268	Fractal path lengthening (geometric path stretch); energy loss only artefact	Docs. 041, 268
P15	061, 041	Marked as ΛCDM reference quantity; static universe has no cosmological z	Doc. 061
P16	026, 064, 181	H_0 is emergent (decompactified), not fundamental	Doc. 263; 026/064/181 flagged
P17	028	DE/DM relation circular; MOND partially derived	Doc. 028 (addendum)
P18	267	Both pictures empirically indistinguishable; decision is theoretical	Doc. 267
P20	267, 268	Form: dimensionless ξ-ratio (R_H/ℓ_P) is fixed	Docs. 267, 268
P29	268	Sequence $\{1, 6, 14, 26\}$ read from data, not derived forward	Doc. 268, open **[S]**
P30	268	Naive spherical Bessel sum does not reproduce $\{3, 18, 42, 78\}$	Doc. 268, open **[S]**
P31	268	T^4 gives harmonic backbone $1 : 2 : 3 : 4 : 5$ (symmetric resonance $k^2 = 3n^2$)	Doc. 268 **[B]**
P32	corpus-wide	H_0 is an imported external calibration, not derived	Doc. 309

No.	Concerns	Subject	Implemented in
P33	009, 042, 181	Form of ξ forward: harmonic $4/3$, spatial dimensions 3, $2^2 = 4$ from geometry	Doc. 009
P34	181, 013	3+1 is empirical input; earlier "follows necessarily" claim overstated	Doc. 181
P35	corpus-wide	ξ^N triviality: "$X = \xi^N$" is meaningless on its own	Docs. 182, 250 (addenda)
P36	182, cosm. sector	$R_H = c/H_0$ is the Hubble length, not the universe size	Doc. 182
P37	009, 188, 271, 272, 275	Three levels clearly separated: micro/meso/macro	respective docs
P38	275	$1/\varphi$ is a transit point, not a fixed point	Doc. 275
P39	182, 267, 277, corpus-wide	H_0/R_H status: no framework derives from first principles	Doc. 309
P40	005, 006, 011, 012, 133, 275	Only ratios exact; absolute values carry fractal/SI correction via bridge constants	corpus-wide
P41	267, 280, 221	Checked, not booked (degeneracy risk)	Doc. 267
P42	282, 292	One declared reference point: m_μ/m_e	Doc. 292
P43	006, 011, 292	K_{frak} not explicit in `calc_En.py`	note
P44	253, 190	ξ as linewidth: disproved for this Shor setup; window width is a measurement-setup property	Doc. 253; Archive-2
R41	284 (cf. 270, 249)	Type-I decompactification is mode-preserving embedding (lossless)	Doc. 270 **[B]**
R42	270 (cf. 248)	Time is relational; symmetry breaking is the measurement process	Doc. 270 **[B]**
R43	285, 283	$C_3 < A_5$, $T^7 = T^4 \times T^3$: HLV $\supset$ FFGFT at group/dimension level	Doc. 285 **[B]**
R44	272, 285	φ acquires meaning as eigenvalue in HLV 5-fold/perp sector, not in FFGFT	Doc. 272 **[B]**
R45	285, 282–284	Quasicrystal: structure/diffraction spectrum point-like; dynamical spectrum continuous	Doc. 285 **[B]**
R46	296 (cf. 025, 026, 063, 156)	Open test point: intermediate-scale formulation cross-cuts corpus line	open **[S]**
R47	285, 297, 298	Spiral-time window not additive but internal (bounded excerpt)	Doc. 285 **[B]**
R48	304 (cf. 295, 301, 303)	Overstatement was in the mail only; document was correct	Doc. 304

No.	Concerns	Subject	Implemented in
R49	304, 305	Consolidated first-mention footnote for U1/U2/U3 and M_τ/M_B (Krüger 2026)	Docs. 304, 305
R50	025, 063, 061, 064, 078	Justification replaced; conclusion (no singular beginning; finite core L_0) unchanged	Doc. 306
R51	049 (cf. 095, 306, 267, 302)	T_0 never denoted time zero, but the smallest possible time interval; notational legacy	Doc. 049 (archived)
R52	101, 165	$L_0 = \xi \ell_P = 2.155 \times 10^{-39}$ m; earlier values corrected	Doc. 180 **[K]**
R53	267 §518	Table row "source of scale": correctly "set / calibrated to H_0"	Doc. 267 **[B]**
R54	292, 306	$N_0 \approx 38.6$ is a ladder-internal factor, not a fundamental unit; value open	Doc. 292 **[S]**
R55	292	Multiplicative structure: third digit arithmetically forced; evidence width one digit	Doc. 292 **[K]**
R56	188 §	Three foundational assumptions declared: $\tilde{T} \cdot m = 1$, T^4, Z_3	Doc. 188 **[K]**
R57	A015	ξ is a fundamental parameter; proof neither possible nor needed	Doc. A015
R58	A142, A140	Quantum gravity not required; $G = \xi^2/(4m_{\text{char}})$ intrinsic	Doc. A142
R59	A130	SI values scheme-dependent (1-loop); fractal corrections not separable	Doc. A130 **[S]**
R60	A160, A165	2π as circle circumference; $\xi/(2\pi)$ as phase correction per measurement	Docs. A160, A165 **[B]**
R61	174, 175	ξ_{Higgs} selection data-driven; two-spaces doctrine withdrawn	Docs. 174, 175 (addenda)
R62	005 §Baryon example	Anchor $\pi^2/2$, not m_μ; error factor 46.705 identified to $0.07\,\%$	Doc. A155 (candidate)
R63	253 §ξ-resonance heuristic	ξ inert in optimal range; optimisation disabled the filter	Doc. 253
R64	2/python/t0_no_fallback_shore.py	Procedure correct; separation plane too coarse; label wrong	Script corrected
R65	024, 075, 076, 147, 176, 253	No procedure distinguishable from Shor; method claim withdrawn	respective docs
R66	097	$16\pi^3$ counting correct, but not for the reason given in Doc. 097	Doc. 310
R67	A040, A270	Rounding interval leaves $n \in [98.3,\ 102.0]$ open; $n = 100$ not exactly forced	Doc. A040

No.	Concerns	Subject	Implemented in
R68	279, 308	$\Lambda^* := 1/R_H^2$ as carrier of the Λ function; name discipline like a_0	Doc. 308
R69	308 §a_0 derivation	Compact time cycle $\tau_c = 2\pi/H_0 = 91.9$ Gyr; energy quantum $\hbar H_0$	Doc. 308
R70	061, A085	T_0 scale gap and notation collision	Docs. 314, 315
R71	309	Sharpened: side-choice in the Hubble tension	Doc. 309 (addendum)
R72	A010, A080, A265	E_0: bare and corrected	Docs. 314, 315
R73	026, 279	H_0 notation, exponent $41/4$	Docs. 026, 279
R74	025, 063	Lithium problem: mechanism rather than solution	Docs. 025, 063
R75	024, 034, 075, 076, 147, 176	Name collision ξ; new edition of Shor documents	Doc. 075 and respective docs
R76	318	Validity range of mass derivations	Doc. 318
R77	041, 160, 318	Classification of early coupling-constant formulas	Doc. 041 (addendum)
R78	160, 005	α_s and running coupling	Doc. 160
R79	321, 319, 049	Algebraic derivation of the $SU(3)_c$ gauge group	Docs. 319, 321
R81	323, 321, 160, 320	Weinberg angle at M_Z from ξ	Doc. 323
R82	322, 320	Hilbert space embedding and spectral theory	Doc. 322
R83	319, 321, 323, 322, 320	Status update of closed bridges	respective docs
R84	324, 321, 009	Casimir derivation of ξ; ξ not a G(6) spectral value	Doc. 324 **[K]**
R85	313, 325, 257, 321, 322	Hawking mechanism microscopically closed	Doc. 325 **[B]**
R86	180, 322, 314, 321, 325	Self-adjointness $\hat{F}$ complete; $\xi = \lambda_{\text{min}}(\hat{F}_{D_4})$	Doc. 327 **[B]**
R87	320, 322	Δm^2_{32}: numerical values corrected ($+16.0\,\%$); closed by Doc. 340 ($1.0\,\%$)	Docs. 320, 340 **[K]**
R88	326, 329	n_{thresh} scale-dependent; Matzke scale $= 2\pi\ell_P/\ln 2$; universal number negative	Doc. 329 **[B]**
R89	180, 019, 201, 032	$m_T = M_{\text{basis}}/\xi$: one structure, sector-dependent basis mass	Doc. 180 **[B]**
R90	180, 019, 201	Exponent in $L_0 = \xi^1\ell_P$ derived from mass term, not by analogy	Doc. 180 **[B]**
R91	019, 201	λ in $m_T = \lambda/\xi$ is a basis mass, not the Higgs quartic	Docs. 019, 201 (addenda)
R92	250, 190-old	Doc. 250: ξ/ℓ is coupling g_T, not m_T; Schwarzschild probe for L_0 is an identity	Doc. 250; this new edition

No.	Concerns	Subject	Implemented in
R93	325, 329	$S_{\text{BH}}(M_{\text{coll}}) = 2\pi$ nat $= n_{\text{thresh}}$ (Matzke) in bits; structural constant	Docs. 325, 329 **[B]**
R94	325	$I_{\text{sector}}/I_{\text{thermal}} = \log_2 3$ for all M	Doc. 325 **[B]**
R95	330	Two formulation clarifications; Type-I theorem [sketch]→**[B]** after Krüger review	Doc. 330 **[B]**
R96	330, 332, 270	Pullback: $L^2(S^1_m) \cong \text{Per}_{R_m}(\mathbb{R}_t)$, not $L^2(\mathbb{R}_t)$	Doc. 332 **[B]**
R97	324, 329	Vacuum operator correction (Matzke): V is involution in $\mathbb{Z}_3\mathbb{C}$	Doc. 324 **[B]**
R98	333, 332, 315, 295, 285	Two meanings of "unrolling" clarified; recursion tied to $\tilde{T} \cdot m$	Doc. 333 **[B]**
R99	334, 307, 306, 285, 174	Superposition without time: classical physics and QM inherit tacit instantaneity	Doc. 334 **[K]**
R100	336, 324, 329	Notation shifted to symmetric $\mathbb{Z}_3\mathbb{C}$; Frobenius fixed points = FFGFT sectors	Doc. 336 **[B]**
R101	317, 336, 338	$\xi = 4/30000$ as Galois number; core identity 43200 from GF$(9)^*$, GF(27)	Docs. 317, 338 **[K]**
R102	338	$1/\alpha = 3700/27 = 137.037$ (7.6 ppm) purely from Galois group orders	Doc. 338 **[K]**
R103	339, 340	Frobenius decomposition GF$(27)^*$; neutrino mass hierarchy	Docs. 339, 340 **[K/B]**
R104	006, 338, 319, 340	$v = 248.3$ GeV $(0.85\,\%)$; G_F, τ_n, Δm derived	Doc. 319 **[K]**
R105	006, 011, 070, 182, 231, 257, 285, 306, 307	Retroactive marker certification (before marker system, 1 Sept. 2026)	Doc. 006 (addendum)
R106	342, 343	Prime forcing by GF(3^k); physical primes $\{5, 7, 11, 13\}$ forced	Doc. 342 **[B]**
R107	343, 338, 162, 186, 328	ξ as resolution floor; Farey coupling; resonance factoriser	Doc. 343 **[B/S]**
R108	340, 342, 343	Dirac/Majorana decomposition of GF$(27)^*$ classes; ν_3 = Dirac neutrino	Docs. 343, 340 **[B]**
R109	006	Neutrino part of Doc. 006 not viable **[X]**; Yukawa method valid for charged leptons	Doc. 006 (addendum)
R110	320, 022	Addendum boxes: ν_3 Dirac neutrino; ξ^3 suppression not a sterile signal	Docs. 320, 022
R111	344–348	Charge quantisation, Gell-Mann-Nishijima and generations from GF$(27)^*$	Docs. 344–348 **[B]**

No.	Concerns	Subject	Implemented in
R112	006, 319, 351	$v = 246$ GeV is SM definition, not a measurement **[Q]**	Doc. 351 **[K]**
R113	338, 317, 011, 351	Anchor m_μ/m_e is QED-dependent; comparison values not theory-free **[Q]**	Doc. 351 **[K]**
R114	351, 352	Lepton residuals: Koide most precise ($0.046 \cdot \xi$); Galois residual $-77.6 \cdot \xi$	Doc. 352 **[K]**
K-041-a	041 §δ_{CKM}	Formula arcsin$\left(2\sqrt{2} \cdot \sqrt{\xi}/3\right)$ gives 0.011 rad; table value 1.20 rad; candidate: arcsin$\left(2\sqrt{2}/3\right)$	Doc. 041 (addendum)
K-041-b	041 §θ_{13}	Formula arcsin$\left(\xi^{1/3}\right)$ gives $2.93°$; table value $8.57°$; candidate: $1/300 = 25\xi$	Doc. 041 (addendum)
K-041-c	041 §$\lvert V_{us}\rvert$	Formula gives 0.2124; table value 0.22452 ($\Delta = 5.4\,\%$)	Doc. 041 (addendum)

.2 Note on R92 (self-correction)

R92(ii) concerned a statement in the *old* Doc. 190 (edition 23 August 2026, now Archive-2), Precision 4: the statement classified there as an internal consistency check — that an object with $M_0 = \xi m_P/2$ would have Schwarzschild radius exactly L_0 — is an *identity* (from $r_s = 2GM/c^2$ and $\ell_P = Gm_P/c^2$ it follows that $r_s(x m_P/2) = x\ell_P$ for every x). It tests neither ξ nor the exponent in L_0. The classification as "probe" was too strong. This finding is resolved in the present new edition; the Archive-2 edition documents the origin.

.3 Currently Open Bridges

- τ_μ directly from FFGFT, without detour via G_F and v (R112, Doc. 351) **[S]**
- Why do ε_i decrease with generation? Attribution to QED extraction + SI projection (R113-bridge, Doc. 352 §10) **[S]**
- Koide amplitude $d/c = \sqrt{2}$ from FFGFT geometry (Doc. 352 §8) **[S]**
- K-041-a/b/c: candidate formulas to be clarified by author review
- Cosmic exponent $41/4$ to be derived forward (P20)
- CMB peak selection: $\{1, 6, 14, 26\}$ to be generated forward (P29/P31)
- C_ℓ projection: physical source/window function (P30)
- Ω_{DM} quantitatively model-neutral (P17)
- Spectral dimension $d_s = 1.86$ vs. FFGFT ≈ 2
- Basis mass $M_{\text{basis}} = m_P$ in the fundamental sector: independent justification needed (R90) **[S]**

- Adversarial D_4 operator specificity test against fitted comparison carriers (R95) **[S]**
- ι-embedding HLV $\subset$ FFGFT in the time sector (Doc. 297)
- R91: label correction $\lambda \to M_{\text{basis}}$ in Docs. 019/201 pending **[S]**
- R46: intermediate-scale formulation (Doc. 296) cross-cuts corpus line; not yet addressed **[S]**